

ACTIVITY

89, 18, 49, 58, 69 $\Rightarrow m=10$

$2k+3\%10$

0	58
1	89
2	49
3	69
4	
5	
6	
7	
8	
9	18

key	Location
89	$18\%10=1$
18	$36\%10=6$
49	$98\%10=8$
58	$158\%10=8$
69	$141\%10=1$

$$\begin{array}{r} 89 \\ 2 \\ \hline 178 \end{array}$$

$$\begin{array}{r} 49 \\ 2 \\ \hline 98 \end{array}$$

$$\begin{array}{r} 58 \\ 2 \\ \hline 116 \end{array}$$

$$\begin{array}{r} 69 \\ 2 \\ \hline 138 \end{array}$$

Quadratic Probing:

0	58
1	89
2	49
3	69
4	
5	
6	
7	
8	
9	18

key	Location	Probing
89	1	1
18	9	1
49	1	2
58	9	2
69	1	3

$(u+i^2)\%M$

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key 49

$i=0 \quad (1+0)\%10=1$

$i=1 \quad (1+1)\%10=2$

key 58

$i=0 \quad (9+0)\%10=9$

$i=1 \quad (9+1)\%10=0$

key 69

$i=0 \quad (1+0)\%10=1$

$i=1 \quad (1+1)\%10=2$

$i=2 \quad (1+2)\%10=3$

Double Hashing:

		key	location	Probing
0				
1	89	89	1	1
2				
3		18	9	1
4	58	49	1 → 7	3
5	69			
6		58	9 → 4	2
7	49	69	1 → 5	4
8				
9	18			

key 49

$$v = (3k + 1) \% 10$$

$$= (3 \times 49 + 1) \% 10$$

$$= 148 \% 10 = 8$$

$$\begin{array}{r} 49 \\ 3 \times \\ \hline 147 \end{array}$$

$$(u + v \times i) \% m$$

$$i = 0$$

$$(1 + 8 \times 0) \% 10 = 1$$

$$i = 1 \quad (1 + 8 \times 1) \% 10 = 9$$

$$i = 2 \quad (1 + 8 \times 2) \% 10 = 7$$

$$\begin{array}{r} 49 \\ 3 \\ \hline 147 \end{array}$$

key 58

$$v = (3k + 1) \% 10$$

$$= (3 \times 58 + 1) \% 10 = 175 \% 10 = 5$$

$$(u + v \times i) \% m$$

$$i = 0 \quad (9 + 5 \times 0) \% 10 = 9$$

$$i = 1 \quad (9 + 5 \times 1) \% 10 = 14 \% 10 = 4$$

key 69

$$v = (3k + 1) \% 10$$

$$= (3 \times 69 + 1) \% 10$$

$$= 208 \% 10 = 8$$

$$\begin{array}{r} 69 \\ 3 \\ \hline 207 \end{array}$$

$$(u + v \times i) \% m$$

$$i = 0 \quad (1 + 8 \times 0) \% 10 = 1$$

$$i = 1 \quad (1 + 8 \times 1) \% 10 = 9$$

$$i = 2 \quad (1 + 8 \times 2) \% 10 = 7$$

$$i = 3 \quad (1 + 8 \times 3) \% 10 = 5$$